

Research paper

Oxytocin receptor genes moderate BDNF epigenetic methylation by childhood trauma

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ABSTRACT

Objectives: Gene-Environment ($G \times E$) interaction is of increasing importance in understanding the pathophysiology of posttraumatic stress disorder (PTSD). This study investigated the interaction effect of childhood traumatic experience and epigenetic methylation of brain-derived neurotrophic factor (BDNF) and a possible moderating effect of oxytocin receptor (OXTR) gene rs53576.

Methods: Ninety-nine patients with PTSD and 81 healthy controls (HCs) were recruited. Clinical assessments, including the childhood trauma questionnaire (CTQ) and posttraumatic stress disorder Checklist (PCL) were performed. BDNF methylation and OXTR genotyping (A vs. G allele) were conducted through blood sampling. A two-way multivariate analysis and a moderated regression analysis were conducted to investigate the moderating effect of the OXTR gene on the relationship between CTQ and BDNF methylation.

Results: As for the HC group, the interaction effect of the CTQ and OXTR genotype was significant on BDNF methylation, and the moderation model showed that CTQ and OXTR group are significant predictors of BDNF methylation. In the $G \times OXTR$ type, the high CTQ group showed a greater BDNF methylation level. As for the PTSD group, no interaction or moderation effects were found.

Limitations: The present study did not control the dosage, duration of medications, and different trauma types and the assessment of childhood trauma was based on self-report.

Conclusions: These results suggested that childhood traumatic experience showed a significant impact on BDNF methylation, and OXTR genes have a moderating effect on this epigenetic mechanism in people who have experienced the childhood traumatic episodes.

1. Introduction

Childhood trauma has consistently been associated with the etiology and severity of post-traumatic stress disorder (PTSD) as well as other psychological diagnoses (Bendall et al., 2012; Brady and Back, 2012; Yehuda et al., 2001). Higher rates of childhood abuse and maltreatment have been found among individuals with PTSD than among those without (Pratchett and Yehuda, 2011). Early life stress is implicated in adult stress response dynamics, including increased susceptibility to stress (Pervanidou et al., 2020). Further, such a persistent impact on the brain and behavior can lead to potential developmental alterations in genome-wide DNA methylation (Smith et al., 2012; Uddin et al., 2010).

Gene and environment ($G \times E$) interaction research has highlighted

stress-induced methylation among the etiological mechanisms of PTSD (Mehta and Binder, 2012; Miller et al., 2020; Smith et al., 2012; Zhang et al., 2014). The brain-derived neurotrophic factor (BDNF) gene is a strong candidate gene related to the development and severity of PTSD (Rakofsky et al., 2012; Voisey et al., 2019). BDNF gene is involved with neuroplasticity and neurotransmission (Binder and Scharfman, 2004) and BDNF methylation is thought to serve various functions, including mediating the stress response in the central nervous system (CNS) through the hypothalamic-pituitary-adrenal (HPA) axis (Hossack et al., 2020). Previous studies have shown that peripheral BDNF concentration is notably decreased in cases of PTSD (Dell'Osso et al., 2009), and that early exposure to stress may result in higher methylation of BDNF (Roth et al., 2009; Roth and Sweatt, 2011). BDNF methylation is thought to be

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involved in long-term perpetuating effects of early life adversity (Roth et al., 2009; Unternaehrer et al., 2012).

Another candidate gene related to the stress response is the oxytocin receptor (OXTR) (Choi et al., 2019; Gouin et al., 2017; Thompson et al., 2011). A variation in OXTR gene rs53576, one of the most studied single nucleotide polymorphisms (SNPs), has been associated with social behaviors and social anxiety (Bartz et al., 2011; Li et al., 2015; Olff, 2012; Rijlaarsdam et al., 2017; Saphire-Bernstein et al., 2011). The A allele carriers have been connected to alterations in the hypothalamus and amygdala at the morphometric level, followed by several malfunctions in prosocial behavior and stress reactivity (Rodrigues et al., 2009; Tost et al., 2010). On the other hand, the G allele (GG) has been connected to prosocial behaviors, including a tendency to seek social support under stress (Chen et al., 2011; Kim et al., 2010; Saphire-Bernstein et al., 2011). Other findings have reported conflicting results where the A allele is associated with more functional social cognitive ability (Park et al., 2010) or the deleterious impact of the G-allele following early-life traumatic experiences (Olff et al., 2013). These conflicting findings could derive from multiple factors, including differences in samples and methodology.

Previous studies have found associations between the OXTR and childhood trauma and between BDNF and childhood trauma. However, only a few studies have probed the interaction between childhood and BDNF methylation (Roth et al., 2009). In particular, recent BDNF methylation studies using Korean samples did not measure early life adversity, which has been linked to methylation of BDNF promoters among both healthy individuals and patients with psychiatric disorders, including PTSD (Kim et al., 2017) and major depressive disorder (Choi et al., 2015; Na et al., 2016). The present study underscores that the differential attachment patterns in adults and their infants are strongly associated with emotional regulation, and several studies have found associations between the OXTR gene and childhood traumatic experiences. However, to date, no studies have probed the interaction between childhood trauma and BDNF methylation while factoring in the effect of the OXTR gene. This study aimed to examine the relationship between childhood trauma and BDNF methylation and the possible impact of the OXTR gene.

2. Materials and methods

2.1. Participants

A total of 180 participants (99 patients with PTSD and 81 healthy controls) between the ages of 18 and 65 (45.39 ± 13.51) were recruited for the study. Non-clinical individuals were recruited from the local community as healthy controls (HC) through flyers and word of mouth. The patients with PTSD were recruited through advertisements on the hospital bulletin board. The diagnosis of PTSD was based on the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) by a board-certified psychiatrist. Participants were excluded if they had sensory impairments, epilepsy, and brain injury. All participants in the HC group reported no history of or current psychiatric medical treatment or need for clinical mental health services. The traumatic events experienced by PTSD patients were: 65 (65.66%) survivors of life-threatening disasters, accidents, and incidents that involved severe physical and/or psychological traumas; and 24 (24.24%) survivors of physical and/or sexual violence or who have recently left abusive and violent relationships; 10 (10.10%) who witnessed the death of a family member, a spouse, or a close friend due to either accident, suicide, or homicide. After a thorough explanation of the procedure and protocol of the experiment, written informed consent was obtained from all participants according to the procedures of the Institutional Review Board at Inje University Ilsan Paik Hospital (IRB no. 2015-07-025).

2.2. Psychological measures

Childhood trauma was examined using a Korean validated version of the childhood trauma questionnaire (CTQ) consisting of five subscales of various childhood traumas, including emotional abuse, physical abuse, sexual abuse, emotional neglect, physical neglect, and another scale detecting minimization and denial (Yu et al., 2009). The CTQ is composed of 28 items and it is assessed with a 5-point Likert scale, ranging from 1 (“never true”) to 5 (“very often true”).

To examine the severity of PTSD symptoms, the *post-traumatic stress disorder checklist for DSM-5* (PCL-5) was administered to assess the severity of PTSD symptoms of the participants (Oh et al., 2014). The PCL is a well-validated self-report rating scale for assessing symptoms of PTSD in Korea. It consists of 20 items and is rated using a 5-point Likert scale, ranging from 1 (“not at all”) to 5 (“extremely”).

2.3. OXTR gene rs53576 polymorphism

All participants had their blood sampled to extract DNA using NanoDrop® ND-2000 UV-Vis Spectrophotometer. The OXTR rs53576 was positioned on chr3:8762685 (GRCh38.p12). The OXTR gene rs53576 polymorphism is analyzed using the TaqMan SNP Genotyping Assays from MacroGen, Inc. (Seoul, South Korea). Genomic DNA was diluted to 5 ng/μl concentration at 96 well PCR plates. TaqMan SNP Genotyping Assays were obtained from Applied Biosystems. The probes were labeled with FAM or VIC dye at the 5' end and a minor-groove binder and non-fluorescent quencher at the 3' end. PCR was done in 5 μl of a mixture containing 2 μl of a DNA sample, 0.125 μl of each TaqMan™ SNP Genotyping Assay (Thermo Fisher Scientific, USA), 2.5 μl of TaqMan™ Genotyping Master Mix (Thermo Fisher Scientific, USA), 0.375 μl of distilled water. Amplification and detection were done with a detection system (QuantStudio 12K Flex Real-Time PCR System, Thermo Fisher Scientific, USA) with the profile of 50 °C for 2 min, 95 °C for 10 min followed by 60 cycles of 95 °C for 15 s and 60 °C for 1 min. After the PCR amplification, allelic discrimination is performed at the same machines (QuantStudio 12 K Flex Real-Time PCR System). The allelic discrimination is an endpoint plate read. The QuantStudio 12 K Flex Software calculates the fluorescence, measurements made during the plate read, and plots Rn values based on the signals from each well. Then analyzed plates were performed through automatic or manual allele calls.

There were three positive control and one negative control sample for each plate. We confirmed the clustering image with positive controls. The intragenomic DNA (gDNA) samples of known genotypes are used for positive control. The OXTR rs53576 was divided into two groups depending on the genotypes: AA + AG (83.81%) and GG (19.19%) (Li et al., 2015).

2.4. BDNF gene pyrosequencing

DNA methylation of a promoter region is generally thought to silence gene expression (Ehrlich and Lacey, 2013; Jones, 2012). Within the BDNF gene, four CpG-sites were selected from Promotor I (chr11: 27744279-27744293) based on previous clinical findings (Choi et al., 2015; Devlin et al., 2010; Keller et al., 2010; Kim et al., 2017; Na et al., 2016). The BDNF promoter region for analyzing methylation status was positioned between −694 and −654 relative to transcription of exon I (CpG1 = −688, CpG2 = −686, CpG3 = −682, and CpG4 = −675) (Kim et al., 2017). DNA was bisulfite modified using EpiTect Fast DNA Bisulfite Kit (Qiagen, Hilden, Germany). Primer sets for PCR amplification and sequencing of methylation target region were designed using the Pyrosequencing Assay Design Software v2.0 (Qiagen). The PCR was performed using Pyromark PCR kit (Qiagen) under the following conditions: 95 °C for 15 min, followed by 45 cycles of 94 °C for 30 s, 58 °C for 30 s, 72 °C for 30 s, then a final extension at 72 °C for 10 min. The BDNF promoter was amplified by PCR according to the operator's

protocol. The amplified PCR products were sequenced using the PyroMark Q48 Autoprep software (Qiagen) for pyrosequencing. The methylation percentage of each CpG site was calculated. Thus, the analysis included the methylation percentages of four individual CpG sites as well as their averaged value.

2.5. Statistical analyses

Chi-square (χ^2) and *t*-tests were performed to determine differences in sociodemographic and clinical assessment variables across PTSD and HC groups. The continuous CTQ scores were dichotomously transformed into low and high CTQ groups based on a median split at 40.00. A series of independent *t*-tests were conducted to see the difference in BDNF methylation between PTSD and HC groups, between high CTQ and low CTQ groups, and between A-OXTR and G-OXTR genotypes. The two-way Analysis of Covariance (ANCOVA) was conducted to examine an interaction effect between CTQ level (Low and High CTQ group) and Oxytocin receptor gene types (A-OXTR carrier and G-OXTR) in terms of BDNF methylation. Age, sex, and years of education were controlled as covariates. Levene's test of homogeneity was carried out to assess the equality of variances across samples. All significant levels were set at $p < .05$ (two-tailed) and all statistical analyses were performed using SPSS (IBM Inc., Chicago, IL, USA).

Subsequently, a moderation regression analysis using Macro PROCES 3.5 for SPSS (version 2.16.3) was performed to examine the moderation effect of OXTR rs53576 on the path between childhood trauma and BDNF methylation. As covariates, demographic factors such as age, sex, and years of education were controlled for their possible association with BDNF methylation and CTQ. Johnson–Neyman technique (Hayes, 2017) was applied to investigate the points in which the moderation variable had a significant effect. The Johnson–Neyman technique continuously aligns the moderation variable and computes the points of significance for interactions by examining the significance between the predictor and outcome variables. A bootstrap method with 5000 replications was applied as an option for moderation regression analysis (Hayes, 2017). To correct for multiple comparison issues, Benjamin and Hochberg's (B-H) procedure was performed for PTSD and HC groups each to test the False Discovery Rate (FDR) test. Thus, a series of partial correlation analyses between CTQ scores and BDNF methylation level with age, sex, and years of education as covariates were performed at each CpG site as well as the mean methylation level were also performed for total participant groups, PTSD only, and HC only.

3. Results

3.1. Descriptive statistics

The HC group was composed of 24 (29.63%) men and 57 (70.37%) women, with a mean age of 46.91 years ($SD = 14.06$). The PTSD group was composed of 33 (33.33%) men and 66 (66.67%) women, with a mean age of 44.13 years ($SD = 12.97$). The years of education, the childhood trauma questionnaire (CTQ), post-traumatic stress disorder checklist (PCL), and hospital anxiety and depression scale (HADS) were significantly different between PTSD patients and healthy participants. The detailed comparisons of demographics, psychological assessments, and genotype frequencies of BDNF and OXTR rs53576 are shown in Table 1.

3.2. Group comparison

Independent *t*-tests showed no significant group difference in BDNF methylation level between PTSD and HC groups across CpG 1 (10.36 ± 3.38 vs. 11.23 ± 4.52 , $t(144.98) = 1.43$, $p = .154$), CpG 2 (9.55 ± 3.31 vs. 10.37 ± 3.97 , $t(155.61) = 1.48$, $p = .141$), CpG 3 (8.43 ± 2.63 vs. 8.98 ± 3.24 , $t(178) = 1.27$, $p = .206$), CpG 4 (11.42 ± 3.43 vs. 12.27 ± 4.10 , $t(178) = 1.52$, $p = .130$), and mean CpG (9.94 ± 3.05 vs. $10.71 \pm$

Table 1

Comparison of the demographic, genetic, and psychological characteristics between participants.

	PTSD (n = 99)	HC (n = 81)	Chi or t	p
	Mean (SD) or N (%)			
Demographic				
Age (years)	44.14 (12.97)	46.91 (14.06)	−0.99	0.322
Sex			0.28	0.60
Male	33 (33.33)	24 (29.63)		
Female	66 (66.67)	57 (70.37)		
Education (years)	12.77 (3.20)	13.71 (3.28)	−2.21	0.028
Psychological assessments				
PCL level	38.49 (17.62)	12.30 (9.58)	12.68	<0.0001
HADS – anxiety	11.34 (4.23)	4.85 (2.24)	13.18	<0.0001
HADS – depression	11.24 (3.61)	4.84 (2.29)	14.44	<0.0001
AUDIT	3.52 (2.00)	2.70 (3.33)	1.53	0.127
CTQ				
Emotional neglect	12.90 (6.52)	9.4 (4.56)	4.09	<0.0001
Physical abuse	10.30 (6.05)	7.82 (3.89)	2.96	0.003
Sexual abuse	7.50 (3.92)	6.18 (2.54)	2.52	0.012
Emotional abuse	10.75 (6.02)	6.79 (3.01)	5.56	<0.0001
Physical neglect	9.36 (3.90)	7.99 (2.94)	2.43	0.016
CTQ (total)	50.78 (20.07)	38.18 (11.38)	4.97	<0.0001
Genetic variants				
BDNF methylation				
CpG 1	10.36 (3.38)	11.23 (4.52)	−1.43	
CpG 2	9.55 (3.31)	10.37 (3.97)	−1.51	
CpG 3	8.43 (2.63)	8.98 (3.24)	−1.27	
CpG 4	11.42 (3.43)	12.27 (4.10)	−1.52	
CpG mean	9.94 (3.05)	10.71 (3.78)	−1.52	
OXTR rs53576			1.45	0.23
AA + AG genotypes	86	65		
GG genotypes	13	16		

Note. Categorical variables are represented as percentages and means (standard deviations) are represented as continuous variables.

Abbreviations: PTSD, post-traumatic stress disorder patients; HC, healthy controls; CTQ, CTQ, Childhood Trauma Questionnaire; PCL, Post-traumatic Stress Disorder Checklist Test; CAPS = Clinician Administered PTSD Scale for DSM-5; HADS = Hospital Anxiety and Depression Scale; AUDIT = Alcohol Use Disorder Identification test; BDNF = brain-derived neurotrophic factor; OXTR = oxytocin receptor.

3.78, $t(178) = 1.52$, $p = .131$), between low CTQ and high CTQ groups across CpG 1 (11.12 ± 4.03 vs. 10.10 ± 3.72 , $t(178) = 1.69$, $p = .093$), CpG 2 (10.20 ± 3.70 vs. 9.39 ± 3.48 , $t(178) = 1.44$, $p = .152$), CpG 3 (8.91 ± 2.93 vs. 8.26 ± 2.89 , $t(178) = 1.44$, $p = .153$), CpG 4 (12.11 ± 3.76 vs. 11.24 ± 3.71 , $t(178) = 1.49$, $p = .139$), and mean CpG (10.59 ± 3.42 vs. 9.74 ± 3.35 , $t(178) = 1.59$, $p = .114$) and between A-OXTR and G-OXTR groups across CpG 1 (10.76 ± 4.02 vs. 10.73 ± 3.64 , $t(178) = 0.043$, $p = .973$), CpG 2 (9.87 ± 3.65 vs. 10.18 ± 3.62 , $t(178) = -0.419$, $p = .676$), CpG 3 (8.66 ± 2.94 vs. 8.76 ± 2.88 , $t(178) = -0.171$, $p = .864$), CpG 4 (11.77 ± 3.77 vs. 11.94 ± 3.76 , $t(178) = -0.216$, $p = .830$), and mean CpG (10.26 ± 3.43 vs. 10.40 ± 3.38 , $t(178) = -0.198$, $p = .844$).

3.3. Two-way ANCOVA analysis

As for the PTSD group, the interaction effect of the CTQ and rs53576 on BDNF methylation was not significant on any BDNF methylation sites or the mean BDNF methylation [$F(1, 88) = 0.542$, $p = .464$]. The development of BDNF methylation was not significantly different regardless of the genotype in people with low CTQ (Fig. 1).

As for the HC group, the interaction effect of the CTQ and OXTR rs53576 was significant on CpG 3 [$F(1, 74) = 6.25$, $p = .015$, $\eta^2 = 0.78$], CpG 4 [$F(1, 74) = 8.21$, $p = .005$, $\eta^2 = 0.10$], and mean BDNF methylation [$F(1, 74) = 8.12$, $p = .006$, $\eta^2 = 0.099$]. In CpG 3, CpG 4, and mean BDNF methylation, there was a significant effect of high CTQ

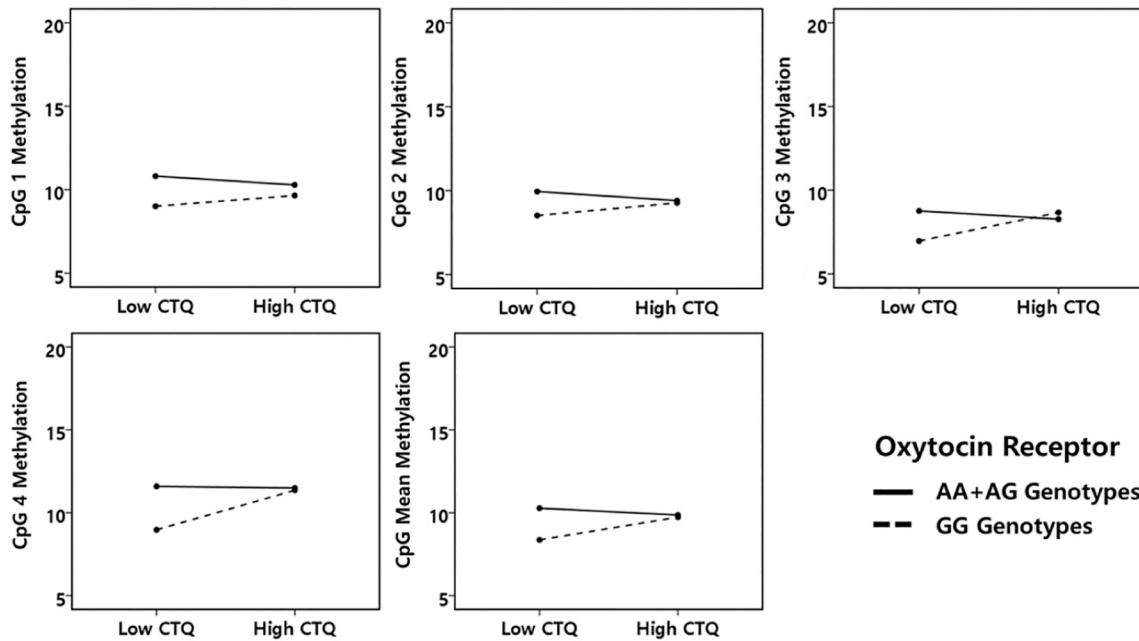
PTSD (*n* = 99)

Fig. 1. PTSD patient group: The two-way ANCOVA analysis of BDNF sites: CpG 1, CpG 2, CpG 3, CpG 4, and CpG mean methylation.

level on BDNF methylation and this effect depended on the difference between A-OXTR and G-OXTR genotype groups (Fig. 2). With high CTQ scores, G allele carriers showed a higher level of BDNF methylation compared to A allele carriers.

As for the total participant group (PTSD + HC), the interaction effect of the CTQ and OXTR rs53567 was not significant on any BDNF methylation sites or the mean BDNF methylation.

3.4. Moderated regression analysis

The moderated regression was executed separately for PTSD and HC patients (Fig. 3). As for the PTSD patients, the overall model confirmed that 32.67% of the variance in the BDNF methylation is being accounted

for in the analysis ($F(6, 74) = 7.4393, p < .0001, R^2 = 0.3267$). The moderation effect was not significant ($F(1, 74) = 0.4615, p = .4986, \Delta R^2 = 0.057$). The model showed that CTQ ($B = -1.2705, t(74) = -0.6691, p = .5051$) and OXTR group ($B = -2.4491, t(74) = -0.9117, p = .3643$) are significant predictors of BDNF methylation. The interaction between CTQ group and OXTR group was not significant ($B = 1.0905, t(74) = -0.9117, p = .3643$).

As for the HC, moderation effects of OXTR rs53567 were statistically significant on the relationship between CTQ and BDNF-mean. The overall model confirmed that 35.26% of the variance in the BDNF methylation is being accounted for in the analysis ($F(6, 74) = 6.7162, p < .0001, R^2 = 0.5938$). The moderation effect was significant and increased by 5.5858% ($F(1, 74) = 6.6101, p = .0120, \Delta R^2 = 0.057$). The

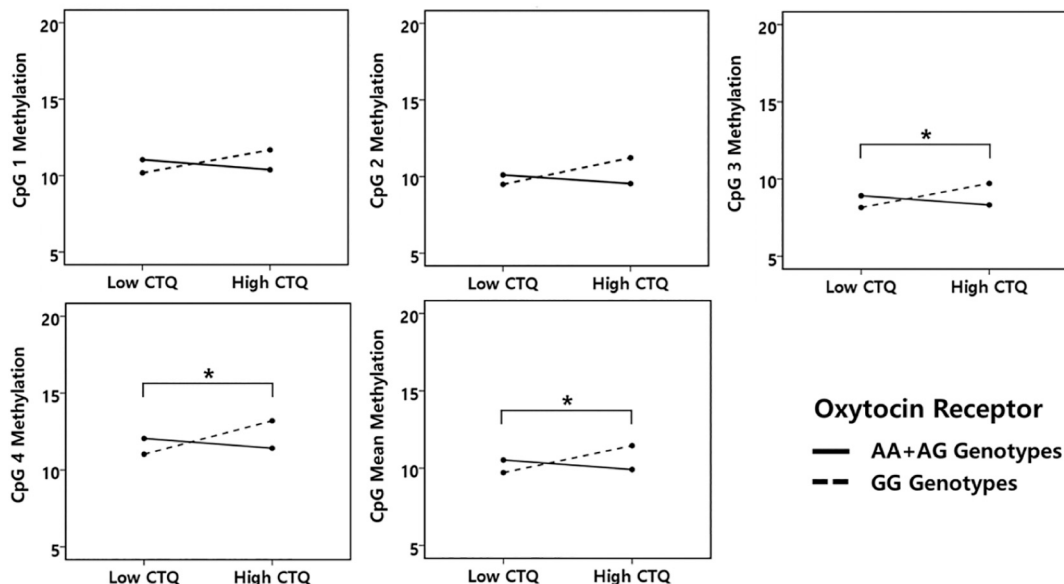
HC (*n* = 81)

Fig. 2. HC group: The two-way ANCOVA analysis of BDNF sites: CpG 1, CpG 2, CpG 3, CpG 4, and CpG mean methylation. * $p < .05$.

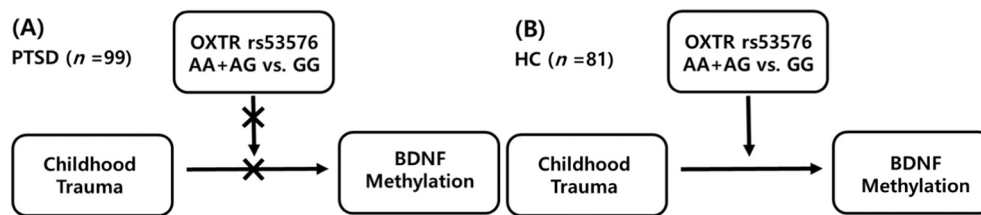


Fig. 3. (A) PTSD patient group: no moderation effect was found. (B) HC group: A moderation of the childhood trauma questionnaire (CTQ) on the effect of oxytocin receptor gene (OXTR) rs53576 on the BDNF (brain-derived neurotrophic factor) methylation level.

model showed that CTQ ($B = -6.6064$, $t(74) = -2.3402$, $p = .022$) and OXTR group ($B = 5.9135$, $t(74) = -2.0095$, $p = .0121$) are significant predictors of BDNF methylation. The interaction between CTQ group and OXTR group was significant ($B = 5.9135$, $t(74) = 2.5710$, $p = .0121$). In the case of A-OXTR group (i.e. AA+AG carriers), there is a non-significant relationship between CTQ and BDNF methylation (Effect = -0.6929 , 95% CI [-2.5937 , 1.2079], $p = .4699$). In contrast, as for G-OXTR group (i.e. G-OXTR carriers), there was a significant positive relationship between CTQ and BD (Effect = 5.2206 , 95% CI [1.0263 , 9.4149], $p = .0154$).

3.5. Partial correlation analysis

Partial correlations between CTQ scores and BDNF methylation level at each CpG site as well as the mean methylation level did not reveal any statistically significant results. As for the total group, no significant correlation was observed across CpG 1 ($r = 0.009$, $p = .097$), CpG 2 ($r = 0.014$, $p = .849$), CpG 3 ($r = 0.042$, $p = .575$), CpG 4 ($r = 0.035$, $p = .645$), and mean CpG ($r = 0.025$, $p = .737$). As for the PTSD group, no significant correlation was shown across CpG 1 ($r = 0.065$, $p = .529$), CpG 2 ($r = 0.040$, $p = .702$), CpG 3 ($r = 0.126$, $p = .221$), CpG 4 ($r = 0.114$, $p = .267$), and mean CpG ($r = 0.090$, $p = .385$). Lastly, within the HC group, no significant correlation was found across CpG 1 ($r = -0.013$, $p = .909$), CpG 2 ($r = 0.035$, $p = .759$), CpG 3 ($r = -0.051$, $p = .655$), CpG 4 ($r = -0.013$, $p = .908$), and mean CpG ($r = -0.009$, $p = .937$).

4. Discussion

In this study, we investigated the interaction effect of traumatic experiences with epigenetic methylation of brain-derived neurotrophic factor (BDNF) and the potential moderating effect of the oxytocin receptor (OXTR) gene rs53576. BDNF methylation was not significantly different between the PTSD and HC groups, between the high CTQ and low CTQ groups, or between A-OXTR carriers and G-OXTR genotypes. A significant moderation effect of OXTR on the relationship between CTQ and BDNF was found only in the HC group. The G-OXTR genotype demonstrated a significant positive relationship between CTQ and BDNF in the HC group.

First, no significant difference in the level of BDNF methylation between the PTSD and HC groups was observed. Previous studies have noted differential epigenetic alterations in BDNF methylation associated with PTSD severity in war veterans (Kim et al., 2017; Voisey et al., 2019). However, this present study did not find evidence of altered BDNF methylation among the PTSD cohort. Small sample size and different types of PTSD triggering traumatic events (e.g., this cohort included victims of traffic accidents and North Korean defectors, but no combat-exposed veterans) could have resulted in these non-significant findings.

Second, a moderating effect of OXTR on the relationship between CTQ and BDNF was found. OXTR has been implicated in a broad range of social and emotional functioning (Lucas-Thompson and Holman, 2013). OXTR rs53576, in particular, is believed to reflect a greater physiological reactivity and subjective sensitivity to environmental stress

depending on the G or A allele carriers (Chagnon et al., 2015). G allele carriers have been connected with prosocial tendencies, while A allele carriers have been linked to malfunction (Chen et al., 2011; Kim et al., 2010; Sapphire-Bernstein et al., 2011). The present study supports this finding. However, in other studies, the A allele has been associated with a more prosocial tendency (Park et al., 2010) and G-allele pertaining to psychopathological development following traumatic experiences (Bradley and Woolley, 2017). The inconsistent findings on the genotype function of OXTR rs53576 suggest possible interference by other unidentified factors (Chang et al., 2014; Krueger et al., 2012), such as early life traumatic experience. Studies have reported that early life traumatic experiences have an adverse impact on mental health in later developmental stages (Kundakovic and Champagne, 2015) and may increase the risk of PTSD (Voisey et al., 2019; Yehuda et al., 2011) with perpetuating consequences due to gene and environmental interaction (Radley et al., 2011; Roth and Sweatt, 2011). The present finding found that while early life traumatic experience is associated with BDNF methylation, and genotype of OXTR plays a crucial moderating role in their relationship.

Further, the altered BDNF methylation profiles found in the HC in the current study were not observed in the PTSD group who were trauma-exposed and with more severity. This may be due to several reasons. First, a more complicated epigenetic mechanism could be involved with patients with PTSD in relation to early life stress (Sheerin et al., 2017). It is well known that PTSD-diagnosed individuals with an early trauma history suffer from more intense aftermath impact of traumatic experiences compared to stress-exposed HCs (Brewin et al., 2000). Various dynamics of genetic and environmental factors could influence BDNF methylation among patients with PTSD (Cornelis et al., 2010; Koenen et al., 2008). The interaction effects have been constantly reported, yet the underlying mechanisms of such interplay are still to be unraveled (Raabe and Spengler, 2013). Therefore, OXTR rs53576 alone may not explain the alterations in BDNF methylation in our cohort.

Secondly, stressors can alter or modulate other variant regions rather than promoter regions of BDNF (Kuan et al., 2017). The sensitivity of BDNF to the stress response has been consistently noted, but its regional expression has been inconsistent (Kim et al., 2017; Voisey et al., 2019). One of the many challenges associated with DNA methylation studies is to connect BDNF methylation changes in peripheral blood cells with changes in the brain and the target tissue (Zheleznyakova et al., 2016). It is likely that the promoter region I in the current study may not be a crucial methylation site interrupted by PTSD symptoms. Future studies may consider other candidate regions rather than promoter regions. Furthermore, medication may be one of the causative factors leading to the null findings in BDNF methylation in our PTSD cohort. Antidepressants are known to influence the level of BDNF methylation (Tadić et al., 2014; Yehuda et al., 2013).

Lastly, the G-OXTR genotype in OXTR rs53567 among individuals with high CTQ levels was related to greater BDNF methylation levels. The present finding showed that greater severity of childhood traumatic experience and G-OXTR interact in their influence on greater levels of BDNF methylation. Several molecular epigenetics studies have confirmed the role of the OXTR rs53576 genotype in the pathophysiology of PTSD (Lucas-Thompson and Holman, 2013; Sippel et al., 2017).

Carriers of A-OXTR have been associated with impaired social behaviors (Kogan et al., 2011; Riem et al., 2011), while the G-OXTR genotype has been linked to greater sociality (Li et al., 2015). On the other hand, others have reported that the A-allele is associated with more functional social cognitive ability (Park et al., 2010) and the G allele with a less resilient coping style under early life adversity (Bradley and Woolley, 2017) and greater sensitivity to social exclusion (McQuaid et al., 2015). Our results can be interpreted as follows. First, the G-OXTR genotype in HCs may represent resilience to stress and more adaptive personality functioning, which can contribute to recovery from traumatic events (Bonanno and Mancini, 2012) and a lower risk of developing PTSD (Horn et al., 2016). Second, it suggests that, given their altered BDNF methylation profiles, these trauma-exposed HCs may manifest trauma-affected stress regulation and behavioral tendencies that are quite different from individuals without early life adversity (Vinkers et al., 2015). Although the precise mechanisms behind it are yet to be identified, our findings indicate that early life stress may contribute to increased BDNF methylation status and that the type of OXTR allele homozygote is critical in this epigenetic difference.

This study has certain limitations. First, the screening of the HC group as well as the diagnosis of PTSD and other psychiatric disorders (e. g., anxiety disorders) were not conducted in the form of a standardized interview. Second, our analysis of specific CpG sites and subscales of childhood traumatic experience was exploratory in nature. Four CpG sites may not be sufficient to detect the overall effects of methylations. The differential influence of each CTQ subscale in relation to the moderating effect of OXTR rs53576 has been noted (Ebbert et al., 2019), but the present study included an individual's traumatic experience as the only main variable. Third, variables in our PTSD sample, including different types of traumatic events, medications, and other mental health issues such as anxiety, depression, and alcohol use, were not controlled in the present study. Further, our PTSD group included North Korean defectors, who may present unique ethnocultural aspects of PTSD. Lastly, we only examined the epigenetic effect through a cross-sectional study. Given the temporal dynamics of traumatic experience, gene and environment interaction research greatly benefits from factoring in time-sensitive variables such as the level of stress over time since the trauma. Thus, longitudinal studies involving genetic diversity may enhance our comprehensive understanding of PTSD-related epigenetic biomarkers and regulations.

In conclusion, we found a significant interaction effect between traumatic experience and epigenetic methylation of BDNF and a possible moderating effect of the OXTR gene rs53576 in the HC. Our results suggest that the oxytocin gene significantly impacts the epigenetic mechanisms of those who have experienced traumatic episodes. Further, it supports that peripheral blood-based BDNF DNA methylation may represent a plausible biomarker for detecting early life adversity.

Data statement

The authors agree to share and make data open to the extent this does not compromise the terms of participant consent and other subject privacy concerns. The detailed version of data is available upon reasonable request to the corresponding author.

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Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Contributors

H.S.L. was responsible for the critical writing and editing of the manuscript. H.S.L. and A.R.K. conducted all major statistical analyses and drafting of the manuscript. S.H.L. and H.S.L. were responsible for conceptualization, design, and editing of the study. S.H.L. supervised the study protocol, data acquisition, analysis, and editing. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare that they have no conflicts of interest.

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